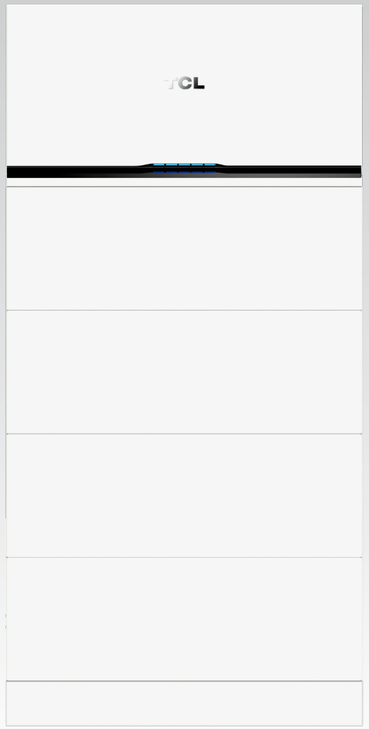
BlueArk X1

3~8kW Single Phase | 5~15kW Three Phase



All-in-One Design

- Integrates hybrid inverter, batteries, UPS, and EMS
- into one system—sleek design, easy installation.
- A reserved port for generator connection is provided;

Full Power Range Coverage

- Single-phase/three-phase inverters with a rated power of 3~8kW/5~15kW;
- Battery capacity ranges from 8kWh to 128kWh
- Supports up to 4 inverters in parallel (both on-grid/off-grid).

High Adaptability, Ultimate safety

- IP66 protection degree, indoor and outdoor installation;
- Integrated with heating module, can operate at temperatures below 0℃ ;
- Type II SPD is equipped on both AC and DC sides.

Excellent Performance

- 3 MPPTs, ideal for complex rooftop scenarios;
- The string current is up to 20A, and the DC/AC ratio is up to 2.0;
- 10ms on/off-grid switching time.

Specifications subject to change without notice.

Datasheet	BlueArk X1 8.0 BMH	BlueArk X1 8.0 BSH*
Performance Specification		
Battery Type	LiFePO4	
Cell Capacity(Ah)	314	
Cycle Life	8000	
Tota Energy Capacity(kWh)	8	
Usable Energy Capacity(kWh)	7.2	
Voltage Range (Single Phase System) (V)	20~60	
Voltage Range (Three Phase System) (V)	20~60	
Max. Charging / Discharging Power(W)	4000	
General Data		
Weight	65.5	63.5
Dimensions (W / H / D)(mm)	780*270*260	
Operating Temperature Range(°C)	Charging: -20 ~ 55 Discharging: -20~55	
Relative Humidity Range	5% ~ 95%	
Max. Operating Altitude(m)	3000	
Cooling	Natural Convection	
System ingress protection rating	IP66	
Installation method	Floor standing / Wall-mounted	
Standard Compliance		
Standard	UN38.3, IEC 62619, IEC 62040, EMC	

BlueArk X1 8.0 BSH cannot be used independently, it must be used in conjunction with the BlueArk X1 8.0 BMH;

Datasheet	BlueArk X1 8.0 Battery	BlueArk X1 16.0 Battery	BlueArk X1 32.0 Battery
Number of Battery Modules	1	2	4
System Consists	1*BlueArk X1 8.0 BMH	1*BlueArk X1 8.0 BMH+1*BlueArk X1 8.0 BSH	2*(1*BlueArk X1 8.0 BMH+1*BlueArk X1 8.0 BSH)
Total Energy Capacity(kWh)	8	16	32
Max. Charging / Discharging Current(A)	157	157	314
Max. Charging / Discharging Power(W)*	4000	8000	16000
Weight(kg)*	70	135	265
Dimensions (W / H / D)*	780*368.5*260	780*638.5*260	780*1178.5*260

*((Include the Battery Base))

BlueArk X1 Single Phase Inverter

Specifications subject to change without notice.

Datasheet	3K-SP	3.6K-SP	4K-SP	4.6K-SP	5K-SP	6K-SP	7K-SP	8K-SP
PV Input Data								
Max. Recommended PV Power (Wp)	6000	7200	8000	9200	10000	12000	14000	16000
Max. PV Input Voltage (V)	600V							
Nominal DC input voltage	360V							
Start-up Input Voltage (V)	100							
MPPT Voltage Range (V)	90 ~ 550							
No. of MPPTs/ Strings per MPPT	3/1+1+1						3/2+1+1	
Max. Input Current per MPPT	20+20+20						32+20+20	
Max. Short-circuit Current per MPPT	30+30+30						50+30+30	
Battery Data								
Compatible Battery	BlueArk X1 8.0 Battery, 8-128kWh							
Max.Charging/Discharging Current (A)	157							
Battery Voltage Range (V)	20~60							
Max.Charging/Discharging Power (W)	3000	3600	4000	4600	5000	6000	7000	8000
Grid Data								
Max.AC Input Power From Grid (VA)	4200	5000	5600	6500	7000	8500	10000	11500
Rated AC Output Power (W)	3000	3680	4000	4600	5000	6000	7000	8000
Max. AC Output Apparent Power(VA)	3300	3680	4400	5060	5500	6000	7000	8000
Rated AC Output Current (A)	13	16	17.4	20	21.7	26.1	30.4	34.8
Max. AC Output Current (A)	15.0	16.0	20.0	23.0	25.0	27.3	31.8	36.4
Rated AC Voltage Range (V)	220 / 230 / 240							
Rated Grid Frequency (Hz)	50 / 60							
Power Factor	0.8 leading ~ 0.8 lagging							
THDi	< 3%(Full Load)							
Backup Data								
Rated Output Active Power (W)	3000	3680	4000	4600	5000	6000	7000	8000
Rated Output Apparent Power(VA)	3300	3680	4400	5060	5500	6000	7000	8000
Nominal Output Voltage	220 / 230 / 240							
Nominal Output Frequency	50 / 60							
THDv	< 3%(R Load)							
Switch Time to Emergency Mode (ms)	10							
Generator(Smart Load) Data								
Grid Type	L+N+PE							
Max. AC input Power(W)	4200	5000	5600	6500	7000	8500	10000	11500
Max. AC input Current(A)	19.1	22.7	25.5	29.5	31.8	38.6	45.5	50.0
Efficiency								
Max.Efficiency	97.60%	97.60%	97.80%	97.80%	98.00%	98.00%	98.10%	98.10%
Protection								
PV Reverse Polarity Protection	Yes							
PV Switch	Yes							
AC/PV Surge Protection	Type II / Type II							
Insulation Resistance Monitoring	Yes							
AC Short-circuit Protection	Yes							
Ground Fault Monitoring	Yes							
Grid Monitoring	Yes							
Anti-islanding Protection	Yes							
Residual-current Monitoring Unit	Yes							
String Monitoring	Yes							
LVRT	Opt							
AFCI	Opt							
General Data								
Dimensions (W / H / D)(mm)	780*400*260							
Weight(kg)	30.5							
Operating Temperature Range(°C)	-30 ~ 60(Derating above 45°C)							
Relative Humidity Range	0% ~ 95%							
Max. Operating Altitude(m)	3000(Derating above 2000m)							
Cooling	Intelligent air cooling							
Protection Degree	IP66							
Communication(WLAN / Ethernet / RS485 / 4G)	Opt							
Warranty(5years/10years)	Yes/Opt							
Standard Compliance								
Standard	IEC/EN 62109-1/2, EN50549-10, AS4777.1, IEC 61727, IEC62116,C10/C11,N4105, IEC 61000, CEI 0-21, IEC61683							

BlueArk X1 Three Phase Inverter

Specifications subject to change without notice.

Datasheet	5K-TP	6K-TP	8K-TP	10K-TP	12K-TP	15K-TP
PV Input Data						
Max. Recommended PV Power (Wp)	10000	12000	16000	20000	24000	30000
Max. PV Input Voltage (V)	1000					
Nominal DC input voltage	700					
Start-up Input Voltage (V)	144					
MPPT Voltage Range (V)	160-1000					
No. of MPPTs/ Strings per MPPT	3/1+1+1				3/2+1+1	
Max. Input Current per MPPT(A)	20+20+20				32+20+20	
Max. Short-circuit Current per MPPT(A)	30+30+30				50+30+30	
Battery Data						
Compatible Battery	BlueArk X1 8.0 Battery, 8-128kWh					
Max.Charging/Discharging Current (A)	314					
Battery Voltage Range (V)	20~60					
Max.Charging/Discharging Power (W)	5000	6000	8000	10000	12000	15000
Grid Data						
Max.AC Input Power From Grid (VA)	7000	8500	11500	14000	17000	22000
Rated AC Output Power (W)	5000	6000	8000	10000	12000	15000
Max. AC Apparent Power (VA)	5500	6600	8800	11000	13200	15000
Rated AC Output Current(A)	7.3	8.7	11.5	14.4	17.3	21.7
Max.AC Output Current(A)	8.3	10.0	13.3	16.7	20.0	21.7
Rated AC Voltage Range (V)	220/380;230/400					
Rated Grid Frequency (Hz)	50 / 60					
Power Factor	0.8 leading ~ 0.8 lagging					
THDi	< 3%					
Backup Data						
Rated Output Active Power (W)	5000	6000	8000	10000	12000	15000
Rated Output Apparent Power(VA)	5500	6600	8800	11000	13200	15000
Nominal output voltage(V)	220/380;230/400					
Nominal output Frequency(Hz)	50 / 60					
THDv	< 3%					
Switch Time to Emergency Mode (ms)	10					
Generator(Smart Load) Data						
Grid Type	3W+N+PE					
Max. AC input Power(W)	7000	8500	11500	14000	17000	22000
Max. AC input Current(A)	10.6	12.9	17.5	21.3	25.8	33.4
Efficiency						
Max.Efficiency	97.80%	97.80%	98.00%	98.00%	98.20%	98.20%
Protection						
PV Reverse Polarity Protection	Yes					
PV Switch	Yes					
AC/PV Surge Protection	Type II/ Type II					
Insulation Resistance Monitoring	Yes					
AC Short-circuit Protection	Yes					
Ground Fault Monitoring	Yes					
Grid Monitoring	Yes					
Anti-islanding Protection	Yes					
Residual-current Monitoring Unit	Yes					
String Monitoring	Yes					
LVRT	Opt					
AFCI	Opt					
General Data						
Dimensions (W / H / D)(mm)	780*480*260					
Weight(kg)	42					
Operating Temperature Range(°C)	-30 ~ 60(Derating above 45°C)					
Relative Humidity Range	0% ~ 95%					
Max. Operating Altitude(m)	3000 (Derating above 2000m)					
Cooling	Intelligent air cooling					
Protection Degree	IP66					
Communication(WLAN/Ethernet/RS485/4G)	Opt					
Warranty(5years/10years)	Yes/Opt					
Standard Compliance						
Standard	IEC/EN 62109-1/2, EN50549-1/10, AS4777.1, IEC 61727, IEC 62116,C10/C11,N4105, IEC 61000, CEI 0-21, IEC61683					